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**A3.** Obtain a minimal POS expression for the output (**L**) from the K-map in A2.

|                                                     |  |  |  |  |  |  |  |     |
|-----------------------------------------------------|--|--|--|--|--|--|--|-----|
| List the PIs and EPIs for <b>L'</b> in product form |  |  |  |  |  |  |  | / 7 |
| PIs                                                 |  |  |  |  |  |  |  |     |
| EPIs                                                |  |  |  |  |  |  |  |     |
| Minimal POS expression: <b>L</b> =                  |  |  |  |  |  |  |  | / 5 |

**A4.** Realise the above POS using a minimum number of 2-input NOR gates:

|                                                                                                      |      |
|------------------------------------------------------------------------------------------------------|------|
| Show the circuit diagram clearly. <b>Both true and complement forms of the inputs are available.</b> | / 10 |
|                                                                                                      |      |

**Task B.** The angle converter sends the current angle as Gray code to a speed monitor, which monitors the rotation speed of the lamp. The data is sent once every clock cycle. Bad weather conditions like lightning sometimes interfere with the signals, and very rarely, **at most a single bit might flip during the transmission** from the converter to the speed monitor. You have to design a system for detecting such errors.

**B1.** Assume that at every clock cycle, the angle code is either unchanged, or increases by 1 unit (for ex 001 → 001 or 001 → 011). Design a sequential circuit with **three inputs** (the current angle code **XYZ**), and **three D flip-flops** to store the angle code received in the previous cycle. You may use the following additional components: **three 2-input XOR gates**, **one 3-to-8 decoder**, and **one 4-input OR gate** to generate the **final output z = 1 if error, 0 otherwise**.

|                    |      |
|--------------------|------|
| Draw your circuit: | / 14 |
|                    |      |

**B2.** Redesign the above circuit (from B1), replacing the decoder and 4-input OR gate with one each of: 2-to-1 Multiplexer, 2-input AND and 2-input OR gate. Keep the rest of the circuit unchanged.

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- If you receive either of these, you can jump to the respective state and output 0 (i.e. no error)
- If you receive 10, it can happen only because the encoder sent 00, but  $B_1$  got flipped during transmission  $\Rightarrow$  corrected next state is 00 (with output 1, i.e. error)
- If you receive 11, it can happen only because the encoder sent 01, but  $B_1$  got flipped during transmission  $\Rightarrow$  corrected next state is 01 (with output 1, i.e. error)

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**Q2.** Implement a 1-bit full subtractor using a PLA of minimum size. Take the inputs as P (MSB), Q and R (LSB) and output as B (for Borrow) and D (for difference).

Truth Table

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K-Maps

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|  |  |
|--|--|
|  |  |
|--|--|

Programming Table:

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|       |              | Inputs |   |   | Outputs |   |
|-------|--------------|--------|---|---|---------|---|
| S.No. | Product term | P      | Q | R | B       | D |
|       |              |        |   |   |         |   |
|       |              |        |   |   |         |   |
|       |              |        |   |   |         |   |
|       |              |        |   |   |         |   |
|       |              |        |   |   |         |   |
|       |              |        |   |   |         |   |
|       |              |        |   |   |         |   |

Implementation:

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